Question 1 Consider the functions $f : [0,1] \to [a,b]$ and $g : [0,1] \to [a,b]$ given by $f(x) = x(b-a) + a$ and $g(x) = \frac{x-a}{b-a}$.

$$g(f(x)) = \tfrac{x(b-a)+a-a}{b-a} = \tfrac{x(b-a)}{b-a} = x.$$

Thus, f is invertible, and therefore is a bijection.

Question 2 (a) Consider the function $f : \mathbb{Z}^+ \to \mathbb{Z} \setminus \{0\}$ given by

$$f(n) = \begin{cases} \frac{n}{2} & n \text{ is even} \\ \frac{-1-n}{2} & \text{otherwise} \end{cases}.$$

(b) *Proof.* We show that g is a bijection by constructing an inverse. Consider the function $m : \mathbb{Q}^+ \to \mathbb{Z}^+$ defined by $m(l) = {p_1}^{f^{-1}(a_1)} \cdots {p_r}^{f^{-1}(a_r)}$. We know f^{-1} exists from (a). Composing m and g, we compute

$$m(g(n)) = {p_1}^{f^{-1}(f(a_1))} \cdots {p_r}^{f^{-1}(f(a_r))} = {p_1}^{a_1} \cdots {p_r}^{a_r} = n.$$

Thus, g is bijective. $\square$

(c) We construct the following function:

$$h(x) = \begin{cases} g(f(n)) & \text{if } n \in \mathbb{Z}^+ \\ -g(f(-n)) & \text{if } n \notin \mathbb{Z}^+ \end{cases}.$$

Question 3 *Proof.* Suppose A, B, C, D are sets, and $A \prec C$. For the sake of contradiction, assume that $D \preccurlyeq B$. There are two cases:

($D \approx B$) If $D \approx B$, then by the transitivity of $\approx$, $D \approx A$. However, if this is the case, then we have $D \prec C$ by our assumption that $A \prec C$. This is a contradiction with our initial assumption that $C \approx D$. Thus, $D \not\approx B$.

($D \prec B$) If $D \prec B$, then there exists an injection $D \hookrightarrow B$. By our assumption that $A \approx B$, there is an injection $B \hookrightarrow A$. The composition of injective functions is injective, and thus there is an injection $D \hookrightarrow A$. Thus, $D \preccurlyeq A$. If $D \approx A$, we reach a contradiction from case 1. If $D \prec A$, we have $D \prec A \prec C$, which would contradict our initial assumption that $D \approx C$.

We reach a contradiction in both cases, and so we must have that $B \prec D$. $\square$

Question 4 *Proof.* Suppose a set A is infinite, and that $B \subseteq A$ is finite. Assume for the sake of contradiction that $A \setminus B$ is finite. We note that $A \setminus B \cup B = A$, and that a finite union of finite sets is finite. Thus, $A \setminus B \cup B = A$ is finite—however, this is a contradiction with out assumption that A is infinite. Thus, $A \setminus B$ is infinite. $\square$

Question 5 *Proof.* Suppose a set A is countable, and that $B \subseteq A$ is countable. Assume for the sake of contradicition that $A \backslash B$ is countable. We note that $(A \backslash B) \cup B = A$. By Theorem 7.6 from the notes we have that the union of two countable sets is itself countable. Thus, we must have that A is countable, which is a contradicition to our assumption that A was uncountable. Thus, $A \setminus B$ is uncountable. □

Question 6 *Proof.* Suppose $\mathbb{Z}[x]$ is the set of all polynomials with integer coefficients. Consider $F_n = \{ f \in \mathbb{Z}[x] \mid \deg(f) \leq n \}$.

Lemma 1. $F_n = \{ f \in \mathbb{Z}[x] \mid \deg(f) \leq n \} \approx \mathbb{Z}^{n+1}$.

We construct a bijection $\mathbb{Z}^{n+1} \to F_n$ via $(a_0, a_1, \ldots, a_n) \to a_n x^n \cdots a_1 x + a_0$.

Appealing to Lemma 1, F_n is countable. Then, we can see that $\mathbb{Z}[x] = \bigcup_{n=0}^{\infty} F_n$. Then, $\mathbb{Z}[x]$ is a countable union of countable sets, and is thus countable. □

Question 7 *Proof.* We follow the proof the Cantor-Shröder-Bernstein theorem, and construct a bijection via the progenitor-ancestor method. We notice that the set of elements in A whose progenitor lies in B is $A_B = \{ \frac{1}{4}, \frac{3}{8}, \ldots \} \cup \{ \frac{3}{4}, \frac{5}{8}, \ldots \}$ and that the set of elements B whose elements lie in B is $B_B = A_B \cup \{ 1 \}$. We construct the following function:

$$h(x) = \begin{cases} \frac{2^{n-1}+1}{2^n} & \text{if } x = \frac{2^n+1}{2^{n+1}} \text{ for some } n \in \mathbb{Z}^+ \\ x & \text{otherwise} \end{cases}.$$

□